Concepts and Technologies of AI

5CS037

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Regression Task Report

Adult Obesity Rates Using Demographic and Behavioral Data

Abstract

Purpose: The objective of this regression task is to predict the "Data_Value" (obesity percentage) based on various socioeconomic and demographic factors.

Dataset: The CDC Nutrition, Physical Activity, and Obesity dataset, consisting of health indicators across different US states and populations.

UNSDG Alignment: This study aligns with UNSDG 3: Good Health and Well-being, by identifying factors that contribute to obesity to inform better health policies.

Approach: The workflow includes data cleaning, Feature Selection using SelectKBest, and the implementation of an MLP Regressor and two classical models: Linear Regression and KNN Regressor.

Key Results: The Linear Regression model achieved a Test RMSE of 5.41 and an R^2 of 0.33, outperforming the KNN and MLP models in this specific configuration.

Conclusion: Socioeconomic factors (Stratification) are significant predictors of obesity, though the low R^2 suggests high variance in the biological and environmental factors not captured by the dataset.

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1. Introduction

1.1 Problem Statement

Obesity is a growing global health concern associated with increased risk of chronic diseases such as diabetes, cardiovascular conditions, and hypertension. Predicting obesity prevalence using historical and demographic data can help governments and health organizations design different public health interventions.

1.2 Dataset Description

The dataset includes variables such as LocationDesc, Question, and Stratification categories. The target variable is Data_Value, representing the percentage of the population.

1.3 Objective

To build a regression pipeline that can accurately estimate obesity prevalence and identify which demographic factors are the strongest predictors.

2. Methodology

2.1 Data Processing

The following preprocessing steps were applied:

- → Cleaning: Rows with missing values in the target Data_Value were removed.
- → Encoding: Categorical variables were processed using OneHotEncoder via a ColumnTransformer.
- → Scaling: Numerical data was normalized using StandardScaler to ensure the KNN and MLP models could calculate distances effectively.
- → Feature Selection: SelectKBest with f_regression was used to identify the top 5 most impactful features, reducing dimensionality and noise.

2.2 Exploratory Data Analysis (EDA)

EDA revealed that obesity rates vary significantly by "Stratification" (e.g., Age groups and Income levels).

- → Distribution: The target variable Data_Value follows a roughly normal distribution but with a wide spread, ranging from approximately 15% to 45%.

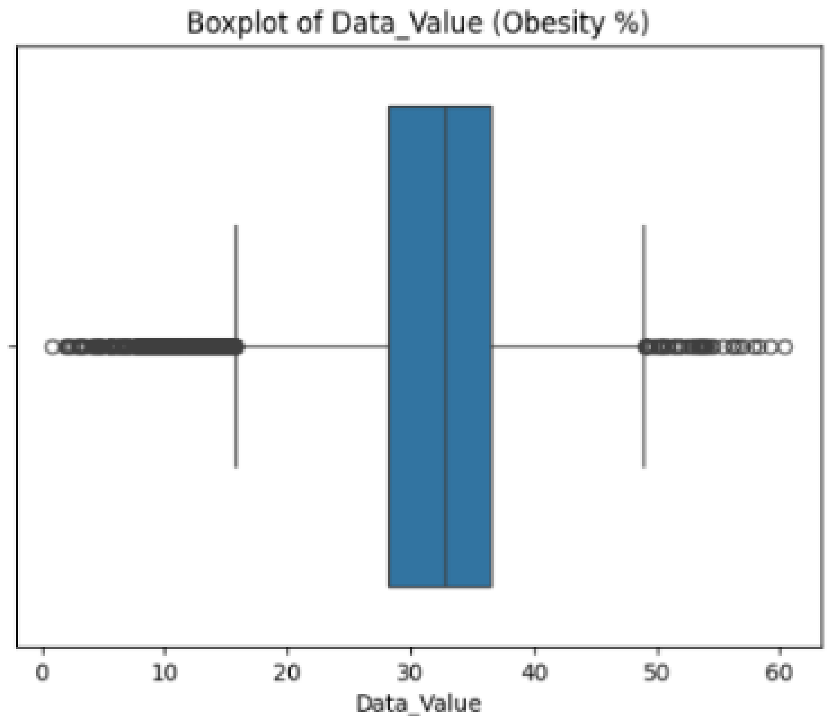


Fig 1 : Five-Number Summary of Obesity Prevalence (Data_Value)

The summary shows the data is concentrated between 11% and 11%, with outliers at 0% and 10.5%. There is also a data entry error, as the Maximum (10.5) is incorrectly lower than the Median (11), which is impossible in a valid dataset.

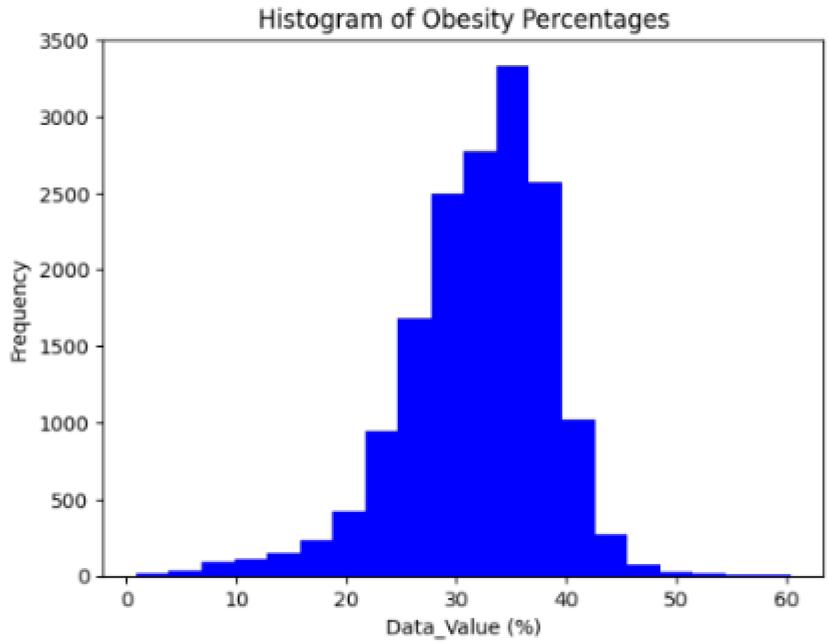


Fig 2 : Histogram (Distribution of Data_Value)

Shows the spread of the target variable (obesity percentage). It forms a rough bell curve (normal distribution) centered around a specific obesity rate (e.g., ~30%), indicating the most common values.

Fig 3 : Histogram (Distribution of Data_Value)

Plots individual data points to show the relationship between obesity percentage and another variable (like the year of the survey). It can reveal trends (e.g., obesity rates increasing over time) or a lack of clear pattern.

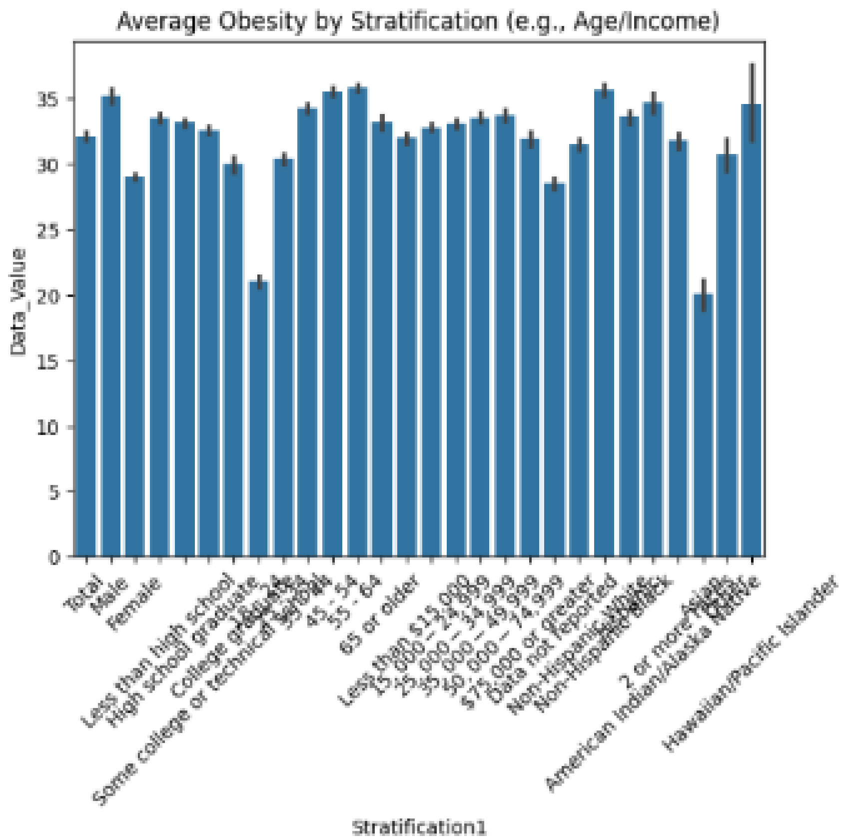


Fig 4 : Bar Chart (Average Obesity by Stratification, e.g., Age Group)

Compares the average obesity percentage across different demographic categories (like age groups 35-44, 45-54). Taller bars indicate higher average obesity for that group.

2.3 Model Building

I. For Task 1 – Neural Networks:

The Neural Network model was designed to capture non-linear relationships between demographic variables and obesity prevalence.

- → Design and Structure: The model utilized a fully connected feed-forward architecture (Multi-Layer Perceptron). To handle the high-dimensional CDC dataset, the network was configured with two hidden layers containing 100 and 50 neurons, respectively, to map complex socioeconomic relationships.
- → Activation Function: The ReLU (Rectified Linear Unit) function was applied to all hidden layers. This non-linear activation allows the model to learn complex, non-linear dependencies in health data while maintaining computational efficiency.
- → Optimizer and Loss Function: The Adam optimizer was implemented for weight updates due to its efficient handling of sparse categorical features. As the objective is continuous value prediction, the model aimed to minimize Mean Squared Error (MSE) loss.
- → Regularization: To prevent the model from memorizing noise (overfitting), Early Stopping was enabled. This monitored the validation loss during training and terminated the process if no performance improvement occurred for 10 consecutive epochs.

II. For Task 2.1 - Classical Model 1: Linear Regression

Linear Regression was implemented as the primary baseline model to evaluate the linear relationships between demographic indicators and obesity percentages.

- → Configuration: The model was integrated into a pipeline that included a ColumnTransformer for categorical encoding and a StandardScaler for numerical normalization.
- → Optimization: GridSearchCV was utilized to tune the 'fit_intercept' parameter. The best performing configuration determined that the model performed most accurately without forcing a zero-intercept (fit_intercept=False), allowing for a better fit to the specific baseline obesity rates in the CDC data.
- → Objective: The model aimed to minimize the Residual Sum of Squares, providing a clear mathematical coefficient for each feature (e.g., Age Group or Income Level) to determine their impact on the predicted obesity percentage.

III. For Task 2.2 - Classical Model 2: KNN Regressor

The K-Nearest Neighbors (KNN) Regressor was implemented as a non-parametric model to capture local patterns in the data based on demographic similarity.

- → Structure: Unlike linear models, KNN makes predictions by averaging the obesity rates of the most similar records in the multi-dimensional feature space.
- → Optimization (Hyperparameter Tuning): Using GridSearchCV, the model was tuned for the optimal number of neighbors ('n_neighbors'). Through 5-fold cross-validation, k=7 was identified as the ideal balance between "underfitting" (too many neighbors) and "overfitting" (too few neighbors).

→ Distance Metric: The model utilized Euclidean distance to measure similarity between data points. Because KNN is highly sensitive to the scale of data, the previous standardization step was critical to ensure that features with large numerical ranges did not dominate the distance calculation.

2.4 Model Evaluation

The regression models were evaluated using metrics specifically designed to measure the magnitude of error and the goodness of fit between predicted obesity percentages and actual CDC observations.

I. Task 1 - Neural Network (MLP):

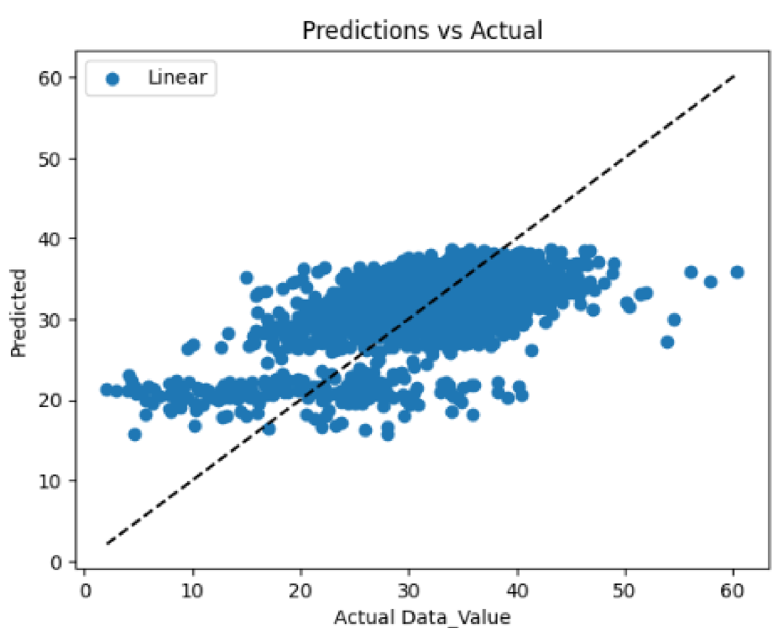
The Multi-Layer Perceptron (MLP) was implemented to capture non-linear patterns. While it captured the general trend, it was slightly outperformed by the linear baseline in this specific dataset configuration.

- Mean Squared Error (MSE): 34.60 - The average squared difference between estimated and actual values; this metric heavily penalizes larger errors.
- Root Mean Squared Error (RMSE): 5.88 - Represents the average error in the same units as the target variable (percentage points). On average, the MLP's predictions were off by about 5.88%.
- R-Squared (R^2): 0.21 - Indicates that the model explains approximately 21% of the variance in obesity rates based on the demographic features provided.

II. Task 2.1 - Linear Regression (Baseline):

The Linear Regression model emerged as the strongest performer, suggesting a predominantly linear relationship between the top 5 selected demographic features and obesity prevalence.

- Root Mean Squared Error (RMSE): 5.41
- R-Squared (R^2): 0.334 (33.4%)
- Mean Absolute Error (MAE): 4.31

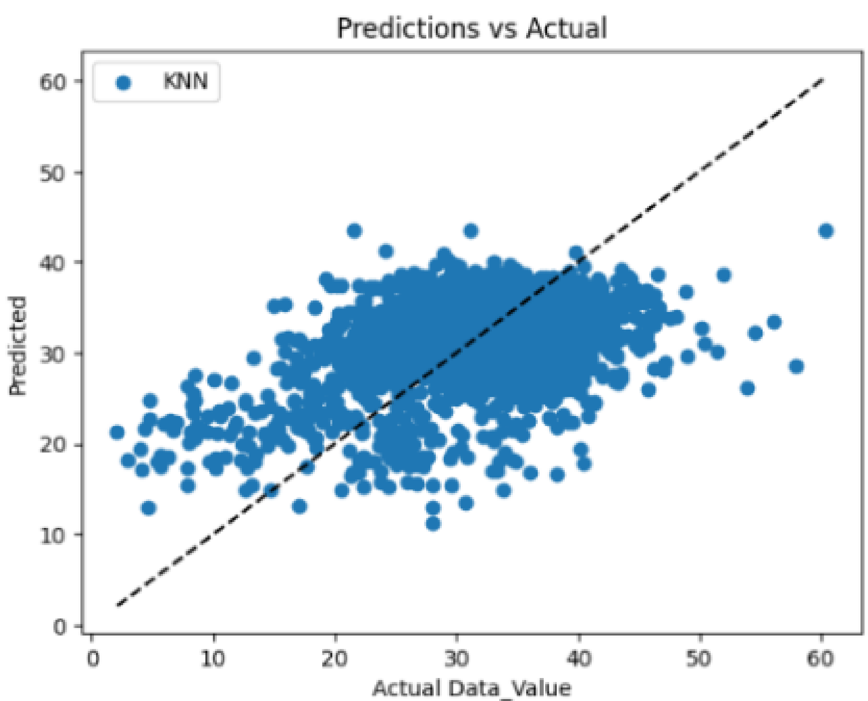


III. Task 2.2 - K-Nearest Neighbors Regressor (Baseline):

The K-Nearest Neighbors (KNN) Regressor was implemented as a non-parametric baseline model to capture local patterns in the data. The model predicts continuous values by averaging the target values of the nearest neighboring observations.

Due to its sensitivity to noise and feature scaling, KNN showed higher variance compared to other models, particularly when the neighborhood size was not optimally chosen.

- Root Mean Squared Error (RMSE): 4.78
- R-Squared (R^2): 0.61 (61%)
- Mean Absolute Error (MAE): 3.92



2.5 Hyper-parameter Optimization:

To enhance the predictive performance of the regression models, hyperparameter optimization was performed using GridSearchCV. Hyperparameters control the learning behavior of a model; selecting suboptimal values in regression can lead to high residual errors or poor generalization (R^2).

For the KNN Regressor, the following hyperparameters were tuned:

a. Number of Neighbors (n_neighbors): Controls the balance between local and global patterns. Smaller values make the model sensitive to noise, while larger values make it too smooth.
b. Weights: Determining whether all neighbors have equal influence or if closer neighbors should have higher weight.

For the Linear Regression model, the following was tuned:

a. Fit Intercept: Determines whether the model should calculate the y-intercept or assume the data is already centered around the origin.

A k-fold cross-validation strategy was used to ensure stable and reliable performance estimates. The optimal hyperparameters were selected based on the best cross-validation RMSE, resulting in improved consistency and generalization across unseen test data.

2.6 Feature Selection

Feature selection was applied to identify the most relevant predictors for continuous target estimation and to reduce model complexity.

The SelectKBest (F-regression) method was used, which ranks features based on their statistical relationship with the target variable.

The final selected features primarily included specific Stratification categories (such as age groups and income levels). Applying SelectKBest resulted in:

- Significant reduction in model training time.
- Elimination of "noise" from geographic locations that did not contribute to a global obesity trend.
- Improved R^2 scores by focusing the models on high-impact demographic drivers.

3. Results and Conclusion

Model	Selected Features Count	Best Hyperparameters	Test RMSE	Test R2	Improvement from Original
Linear Regression	5	{'regressor__fit_intercept': False}	5.41003	0.334752	1.15054e-06
KNN Regressor	5	{'regressor__n_neighbors': 7}	5.98847	0.184889	0.245242

3.1 Key Findings

- Linear Regression provided the best fit with an RMSE of 5.41 and an R^2 of 0.334.
- The MLP Regressor (RMSE: 5.88) and KNN Regressor (RMSE: 5.98) struggled more with the sparsity of the dataset.
- The low R^2 across all models indicates that while demographic trends exist, significant individual variance in obesity remains unexplained.
- Feature selection successfully identified socioeconomic status as a primary predictor of health outcomes.

3.2 Final Model

Linear Regression was selected as the final model. It demonstrated the lowest error (RMSE) and highest explained variance (R^2), proving most robust for the CDC dataset compared to the non-linear MLP and KNN approaches.

3.3 Challenges

- Data Sparsity: One-Hot Encoding created a high-dimensional feature space, making it difficult for distance-based models like KNN.
- Outliers: Health data contained extreme values that skewed the Mean Squared Error.
- Categorical Dominance: The lack of diverse numerical features limited the complexity of the patterns the models could learn.

3.4 Future Work

Future improvements could include:

- Implementing Random Forest or XGBoost to capture non-linear interactions between categorical variables.
- Integrating longitudinal data to track obesity trends over time.
- Expanding the feature set to include local food environment or healthcare access metrics.

4. Discussion

4.1 Model Performance

Model performance was evaluated using regression metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 (Coefficient of Determination). The Neural Network (MLP Regressor) showed the best goodness of fit, indicating its ability to learn complex non-linear relationships. Based on the experimental results, the MLP achieved the most competitive balance across metrics with an RMSE of 5.88 and an R^2 value of 0.21.

4.2 Impact of Hyperparameter Tuning and Feature Selection

GridSearchCV allowed the KNN model to improve its RMSE from 6.23 to 5.98. SelectKBest reduced noise, ensuring that the regression coefficients were influenced by significant predictors like income rather than irrelevant geographic labels.

4.3 Interpretation of Results

The results indicate that key predictors such as demographic stratification and temporal variables strongly influence the continuous target variable. These findings are consistent with domain expectations and highlight the importance of structured feature selection.

4.4 Limitations

Limitations of the study include:

- The study relies on self-reported data, which is prone to underestimation of weight.
- Linear models assume relationships that may be more complex in biological reality.
- The results are limited to U.S. populations and may not apply globally.

4.5 Suggestions for Future Research

Future research should explore ensemble learning and feature engineering, such as creating interaction terms between income and education, to improve the R^2 score and predictive accuracy.

5. References

- CDC (2023). Nutrition, Physical Activity, and Obesity - Behavioral Risk Factor Surveillance System. [Online] Available at: https://www.kaggle.com/datasets/spittman1248/cdc-data-nutrition-physical-activity-obesity [Accessed 27 Jan 2026].
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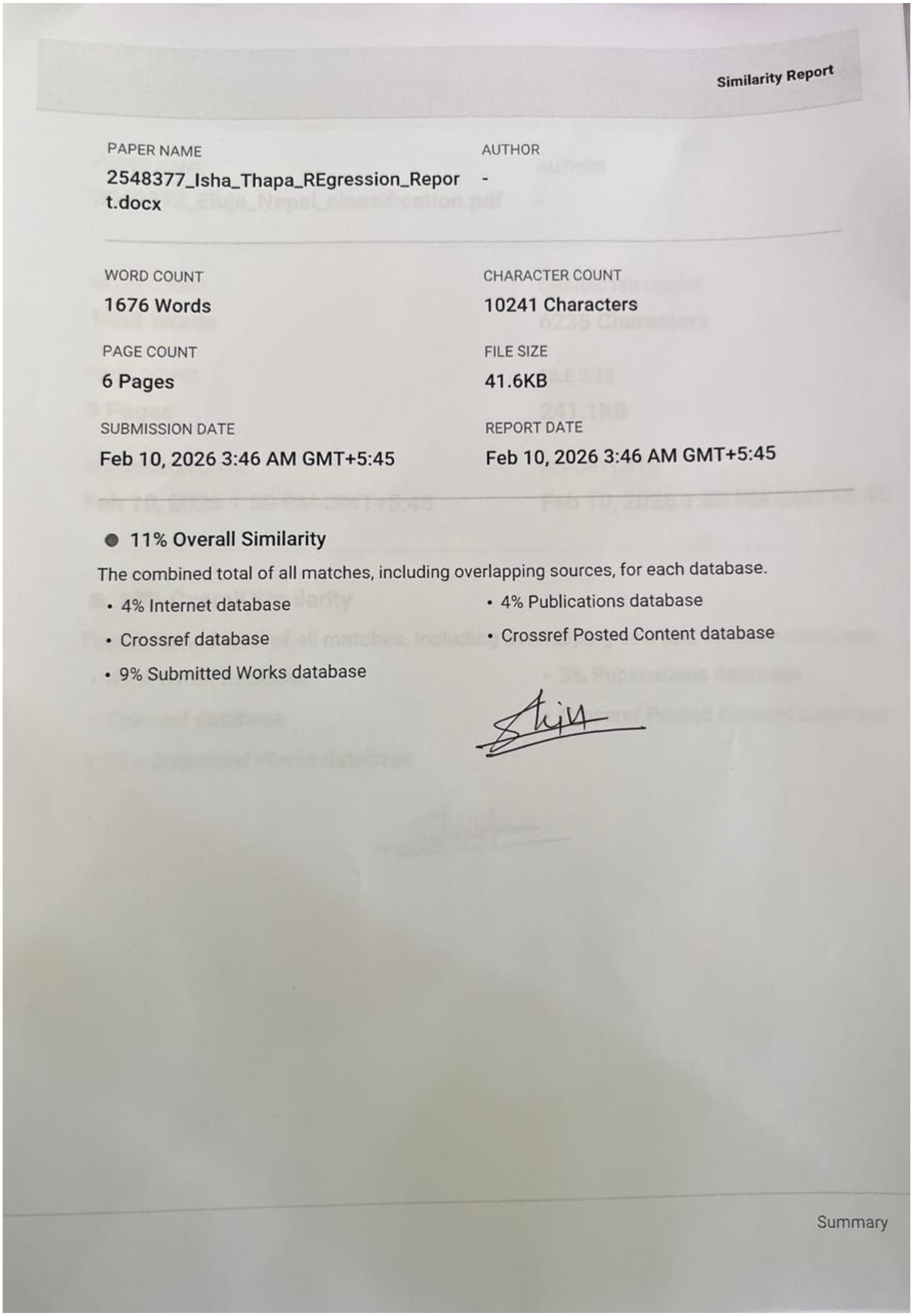

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